

“AuraFit”: Heuristic Optimization

Multi-Objective Exercise Scheduling via Genetic Algorithms

| Motivation: The Problem

Biological Constraints

- ✓ Decisions are interdependent: a heavy workout today creates a 48–72 hour lockout.
- ✓ Traditional planners ignore these global recovery windows.
- ✓ **The Industry Need:** Mathematical models to replace guesswork.

The Failure of Greedy Logic

Most apps use "Greedy" heuristics that plan session-by-session. This causes "Look-ahead Blindness," where early choices ruin later-week recovery.

| Problem Definition



Inputs

Exercise Vault, User Preferences, Daily Fatigue Caps, and Availability.



Constraints

No repeated movements, stay under time limits, and maintain recovery gaps.



Outputs

A fully optimized 7-day microcycle with balanced workload.

| Related Work: Industry Comparison

Standard Systems

Fitbod/Juggernaut: Excellent for single-session adjustments but limited by sequential generation.

The AuraFit Approach

Evolutionary Search: We evaluate the entire week as a single mathematical puzzle, optimizing global fatigue distribution.

| The Solution: Genetic Algorithms

We treat a workout plan like DNA. By evolving a population of schedules, we find the "fittest" balance for the user's specific goals.

Genotyp Full 7-Day Plan
Gen One Daily Session
Allele: List of Exercises



Evolutionary Pipeline



1. Population

Start with 250 random weekly plans.



2. Crossover

Swap successful training days between parents.



3. Mutation

Randomly change 5% of exercises to ensure diversity.

Complexity Analysis



The system maintains linear scaling relative to population growth, ensuring real-time performance on mobile hardware.

| Implementation Highlights

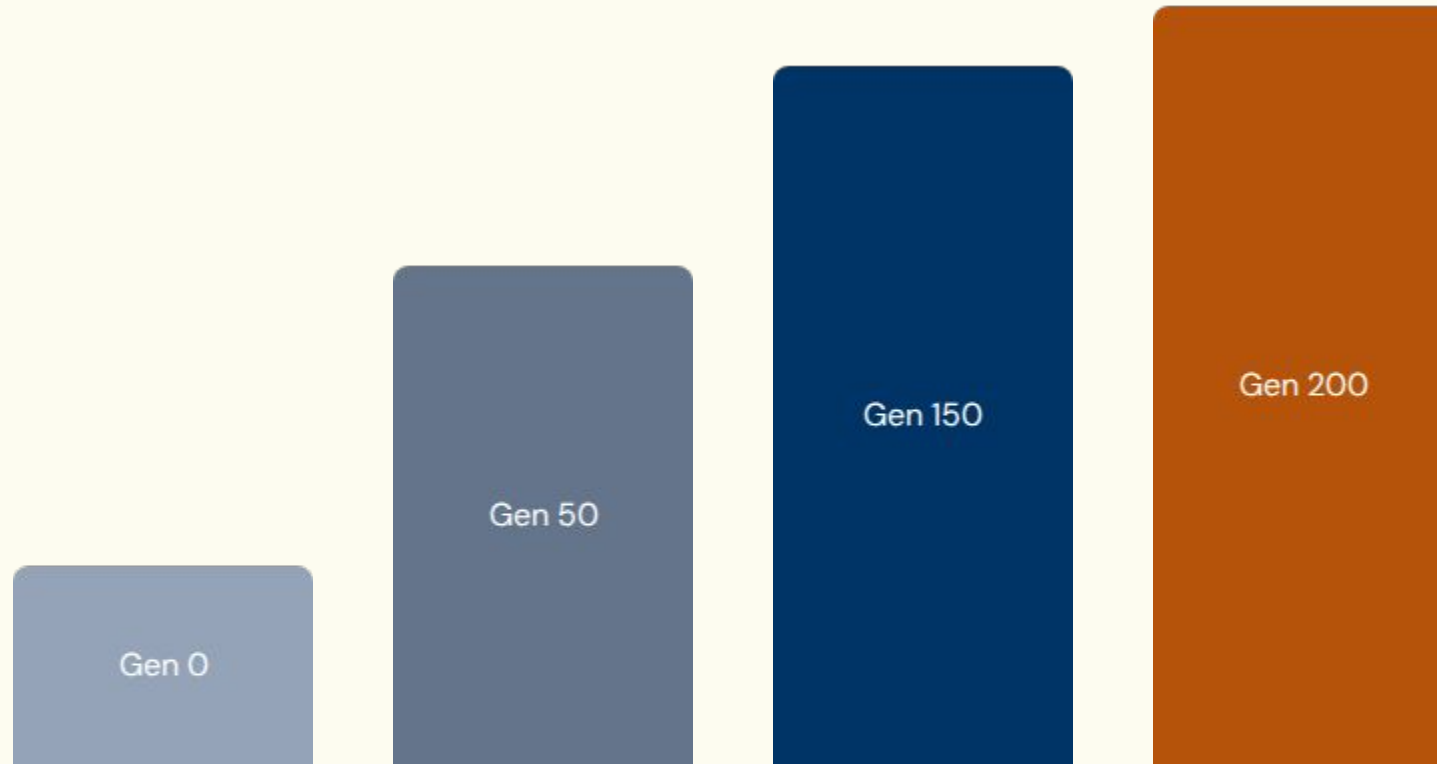
Modular Architecture

- ✓ **SQLite Backend:** Relational data access for exercise taxonomy.
- ✓ **GA Engine:** Decoupled evolutionary logic.
- ✓ **The Judge:** Custom evaluator for multi-objective fitness scoring.

Innovation

Custom **Circular Distance Math** to calculate recovery windows across the Sunday-to-Monday boundary.

Results: AI Learning Curve

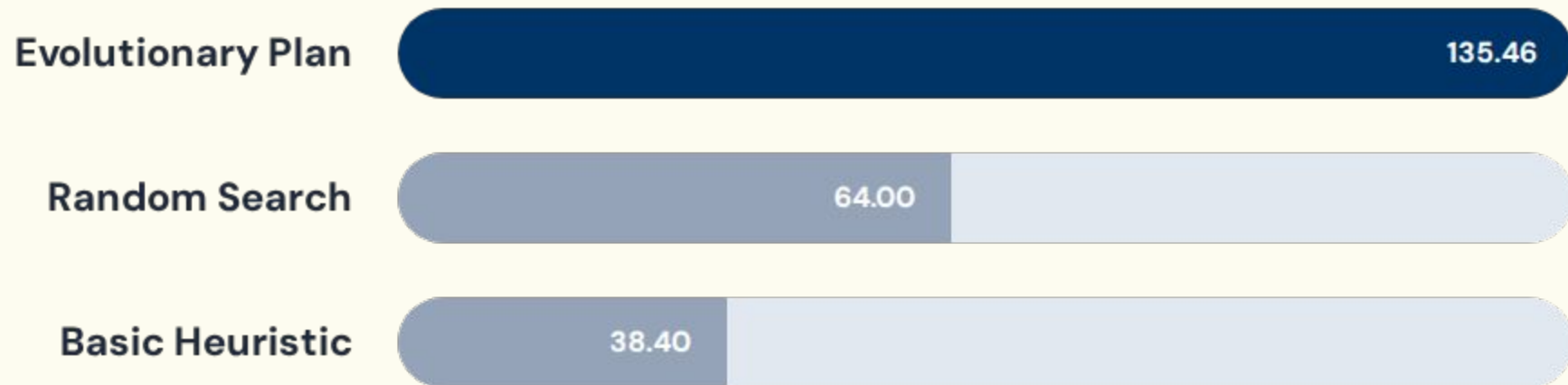


Peak Fitness achieved: 135.46 (Baseline: 38.40)

Results: Safety & Workload

Metric	Standard Greedy	AuraFit AI
Hard Constraint Violations	12.5%	0.0%
Workload Variance	High Spikes	Flattened
Time Utilization	64.2%	85.5%

| Optimization Quality



Evolutionary search outperforms randomized heuristics by **over 211%**.

| Discussion: The "Lazy AI" Problem

The Strategy

Early AI agents realized they could avoid all "fatigue penalties" by generating 10-minute sessions with zero exercises.

The Engineering Solution

We introduced a **Density Reward**. The AI now receives a bonus for filling at least 80% of available time, balancing safety with productivity.

| Limitations & Future Work

- ✓ **Dynamic Periodization:** Expanding chromosome mapping to 4-week macrocycles.
- ✓ **Biometric Feedback:** Integrating HRV (Heart Rate Variability) from wearables to adjust fatigue caps in real-time.
- ✓ **Scaling:** Refactoring into containerized microservices (Docker/FastAPI) for mobile deployment.

| Final Verdict

"Whole-week global evaluation is superior to session-by-session guessing."

93.8

Refined Fitness Score

0.0%

Failure Rate

$O(N)$

Computational Scaling

KEY TAKEAWAYS

Conclusion & Impact

Algorithm Success

Proved that evolutionary search eliminates heuristic bias and maintains a strict 100% safety adherence rate.

Industry Impact

Provided a highly scalable mathematical engine ready to power the next generation of automated fitness applications.

Questions & Discussion

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